Lines, angles, and triangles — Part 2 answer key

#	ID	Answer	Difficulty
2.1	08390-DC	B	Medium
2.2	05a3c218-67ad-4fcc-a95c-61a1756c68c5	C	Medium
2.3	715c4c96-5aac-4b8c-9adb-e5f7c886e5e6	63	Medium
2.4	c39deac6-0f67-4431-b530-e4745624b358	57	Medium
2.5	ec46bc72-04be-4fc9-91f0-f123c8dcce22	C	Medium
2.6	028920-DC	D	Medium
2.7	06625-DC	B	Medium
2.8	c58fbfdc-93b3-4cba-83da-ffc2c85dd427	D	Medium
2.9	2a19ba5d-f615-44cf-bbf1-badb5fc9d6f7	B	Medium
2.10	2cd72fbe-243f-4276-9b51-87f0d4ffa44d	A	Medium
2.11	57d14ecb-ffd5-4286-be93-6ac79a3d7750	B	Medium
2.12	e99677cc-b481-49aa-b1e3-816b48a49b59	B	Medium
2.13	c498bacc-713c-42ba-a2e4-bc873030515a	D	Medium
2.14	009bb623-417f-401a-bef5-57d78eeaaf99	A	Medium
2.15	29e0970f-9765-4c31-bad7-450cfe67c9c9	C	Medium
2.16	86d8b279-9fd7-4afb-97df-f463761e3ebe	B	Medium
2.17	7ad5d0d9-6334-465a-9284-026de00b1ddf	84	Medium
2.18	f821b3cc-a3b5-400b-b813-232bf82351bb	C	Medium
2.19	d1663b7f-c5a4-46e7-b862-f65ceef323c2	54	Hard
2.20	669ffcd9-3468-483d-9398-56885814c9b7	C	Hard

2.21	8cc3132c-df20-42a0-a034-412f01a73b1e	156	Hard
2.22	4e338dff-7649-48c0-89f7-665b5a7d5776	C	Hard
2.23	21c2c201-137b-4fd3-9fa4-520c318b5319	14.66, 14.67, 44/3	Hard
2.24	070630-DC	?	Hard
2.25	dbda3b6a-f820-4919-8708-c6088f04c080	83	Hard
2.26	bcd58635-d61a-4ff2-b368-7bc500780c03	D	Hard
2.27	ba093354-b8dd-457d-94ec-dc6f8b6136ff	B	Hard
2.28	c7393c8f-4256-438b-bd7c-1da38024e64c	153	Hard
2.29	f05a0950-f543-4887-9b76-bb724f91cc50	D	Hard
2.30	fb64be41-9c0d-458c-a301-7cdb0ed92404	54	Hard
2.31	017fc25c-1859-481d-88b6-98502365109e	480	Hard
2.32	022197-DC	C	Hard
2.33	ce3b6f4f-60f0-4503-b733-4d7804f8a14b	C	Hard
2.34	05757-DC	?	Hard
2.35	4c3d0cff-974b-43c0-be0a-e9085cd5448d	A	Hard
2.36	c48a5175-a810-4a78-babf-4e18402eb1a3	C	Hard
2.37	0e784b4f-a1ef-4649-a002-0c63e83b3d22	B	Hard
2.38	33267cea-2b74-4c39-98b4-c53cd17a378d	101/2, 50.5	Hard
2.39	6c28e838-bf2a-4be6-ba95-64b7eb7c520d	D	Hard
2.40	9f3f179f-7ad7-4230-8a40-f348dbb4881a	D	Hard
2.41	ece15db3-71cb-4f5a-8d30-df4238c90943	54	Hard
2.42	536a175a-fa2a-4666-b79d-655eb052f3d9	D	Hard
2.43	11114-DC	C	Hard
2.44	8f2e34ca-40c9-466a-beaf-7b8cda35144d	A	Hard
2.45	8a0d3a20-6326-4fe6-a269-38ef70418956	123	Hard
2.46	070631-DC	?	Hard

2.47	761b928d-225b-4b73-968f-a9b367e5083e	D	Hard
2.48	05759-DC	?	Hard
2.49	23141598-61c1-4eb9-a3fc-9ffb6c807ef1	47	Hard
2.50	3d5eb413-328a-4e40-88e5-debcb6d91aad	1660	Hard
2.51	a8b595d8-c45d-47e6-9ef2-b8a5d7a44e1c	A	Hard

Rationales

▼ 2.1 — B

Choice B is correct. Since $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel, quadrilaterals $AFEB$ and $BEDC$ are similar. Let x represent the length of $\overline{ED}$. With similar figures, the ratios of the lengths of corresponding sides are equal. It follows that $\frac{9}{18.5} = \frac{8.5}{x}$. Multiplying both sides of this equation by 18.5 and by x yields $9x = (18.5)(8.5)$, or $9x = 157.25$. Dividing both sides of this equation by 9 yields $x = 17.47$, which to the nearest tenth is 17.5.

Choices A, C, and D are incorrect and may result from errors made when setting up the proportion.

▼ 2.2 — C

Choice C is correct. The sum of the interior angles of a triangle is $180°$. It's given that the interior angles of triangle ABC are $54°$, $90°$, and $\frac{k}{2}°$. It follows that $54 + 90 + \frac{k}{2} = 180$, or $144 + \frac{k}{2} = 180$. Subtracting 144 from each side of this equation yields $\frac{k}{2} = 36$. Multiplying each side of this equation by 2 yields $k = 72$. Therefore, the value of k is 72.

Choice A is incorrect. This is the value of $\frac{k}{2}$, not k.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.3 — 63

The correct answer is 63. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D. Corresponding angles of congruent triangles are congruent and,

therefore, have equal measure. It's given that the measure of angle A is $27°$. It follows that the measure of angle D is also $27°$. It's given that angle E is a right angle. Therefore, the measure of angle E is $90°$. Let x represent the measure, in degrees, of angle F. Since the measures of the angles in a triangle sum to $180°$, it follows that $27 + 90 + x = 180$, or $117 + x = 180$. Subtracting 117 from both sides of this equation yields $x = 63$. Therefore, the measure, in degrees, of angle F is 63.

▼ 2.4 — 57

The correct answer is 57. Based on the figure, the angle with measure $y°$ and the angle vertical to the angle with measure $58°$ are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure $58°$ also has measure $58°$. It's given that lines q and r are parallel. Therefore, same side interior angles between lines q and r are supplementary. It follows that $y + 58 = 180$. If $y = 2x + 8$, then the value of x can be found by substituting $2x + 8$ for y in the equation $y + 58 = 180$, which yields $2x + 8 + 58 = 180$, or $2x + 66 = 180$. Subtracting 66 from both sides of this equation yields $2x = 114$. Dividing both sides of this equation by 2 yields $x = 57$. Thus, if $y = 2x + 8$, the value of x is 57.

▼ 2.5 — C

Choice C is correct. It's given that $\triangle ABC$ is congruent to $\triangle BDE$. Corresponding sides of congruent triangles are equal in length. It's given that r, which represents the length of $\overline{AB}$, is equal to 54 and s, which represents the length of $\overline{DE}$, is equal to 26. Since $\overline{AB}$ and $\overline{DE}$ aren't equal in length, it follows that $\overline{AB}$ corresponds to $\overline{BD}$ and $\overline{BC}$ corresponds to $\overline{DE}$. Therefore, the length of $\overline{BD}$ is 54 and the length of $\overline{BC}$ is 26. The points B, C, and D lie on a straight line, so the length of $\overline{BD}$ can be expressed as the sum of the lengths of $\overline{BC}$ and $\overline{CD}$. This relationship can be written as $BD = BC + CD$. Substituting 54 for BD and 26 for BC in this equation yields $54 = 26 + CD$. Subtracting 26 from both sides of this equation yields $CD = 28$. Therefore, the length of $\overline{CD}$ is 28.

Choice A is incorrect. This is the length of $\overline{BC}$, not $\overline{CD}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length of $\overline{BD}$, not $\overline{CD}$.

▼ 2.6 — D

Choice D is correct. The sum of the three interior angles in a triangle is $180°$. It's given that angle A measures $50°$. If angle B measured $100°$, the measure of angle C would be $180° - (50° + 100°) = 30°$. Thus, the measures of the angles in the triangle would be $50°$, $100°$, and $30°$. However, an isosceles triangle has two angles of equal measure. Therefore, angle B can't measure $100°$.

Choice A is incorrect. If angle B has measure $50°$, then angle C would measure $180° - (50° + 50°) = 80°$, and $50°$, $50°$, and $80°$ could be the angle measures of an isosceles triangle. Choice B is incorrect. If angle B has measure $65°$, then angle C would measure $180° - (65° + 50°) = 65°$, and $50°$, $65°$, and $65°$ could be the angle measures of an isosceles triangle. Choice C is incorrect. If angle B has measure $80°$, then angle C would measure $180° - (80° + 50°) = 50°$, and $50°$, $80°$, and $50°$ could be the angle measures of an isosceles triangle.

▼ 2.7 — B

Choice B is correct. In right triangle ABC, the measure of angle B must be 58° because the sum of the measure of angle A, which is 32°, and the measure of angle B is 90°. Angle D in the right triangle DEF has measure 58°. Hence, triangles ABC and DEF are similar (by angle-angle similarity). Since $\overline{BC}$ is the side opposite to the angle with measure 32° and AB is the hypotenuse in right triangle ABC, the ratio $\frac{BC}{AB}$ is equal to $\frac{DF}{DE}$.

Alternate approach: The trigonometric ratios can be used to answer this question. In right triangle ABC, the ratio $\frac{BC}{AB} = \sin(32°)$. The angle E in triangle DEF has measure 32° because $M(\angle D) + M(\angle E) = 90°$. In triangle DEF, the ratio $\frac{DF}{DE} = \sin(32°)$. Therefore, $\frac{DF}{DE} = \frac{BC}{AB}$.

Choice A is incorrect because $\frac{DE}{DF}$ is the reciprocal of the ratio $\frac{BC}{AB}$. Choice C is incorrect because $\frac{DF}{EF} = \frac{BC}{AC}$, not $\frac{BC}{AB}$. Choice D is incorrect because $\frac{EF}{DE} = \frac{AC}{AB}$, not $\frac{BC}{AB}$.

▼ 2.8 — D

Choice D is correct. When two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. It's given that in triangle *DEF*, the measure of angle *D* is $47°$ and the measure of angle *E* is $97°$. It's also given that in triangle *RST*, the measure of angle *R* is $47°$ and the measure of angle *S* is $97°$. It follows that angle *D* is congruent to angle *R* and that angle *E* is congruent to angle *S*. Therefore, triangle *DEF* is similar to triangle *RST* and no additional information is needed.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 2.9 — B

Choice B is correct. Each tree and its shadow can be modeled using a right triangle, where the height of the tree and the length of its shadow are the legs of the triangle. At a given point in time, the right triangles formed by two nearby trees and their respective shadows will be similar. Therefore, if the height of the other tree is *x*, in feet, the value of *x* can be calculated by solving the proportional relationship $\frac{10 \text{ feet tall}}{5 \text{ feet long}} = \frac{x \text{ feet tall}}{2 \text{ feet long}}$. This equation is equivalent to $\frac{10}{5} = \frac{x}{2}$, or $2 = \frac{x}{2}$. Multiplying each side of the equation $2 = \frac{x}{2}$ by 2 yields $4 = x$. Therefore, the other tree is 4 feet tall.

Choice A is incorrect and may result from calculating the difference between the lengths of the shadows, rather than the height of the other tree.

Choice C is incorrect and may result from calculating the difference between the height of the 10-foot-tall tree and the length of the shadow of the other tree, rather than calculating the height of the other tree.

Choice D is incorrect and may result from a conceptual or calculation error.

▼ 2.10 — A

Choice A is correct. It's given that triangle *TUV* is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$, where *T* corresponds to T' and *U* corresponds to U'. A dilation is a transformation that multiplies all side lengths by the same scale factor while preserving

angle measures. Therefore, corresponding angles in triangles TUV and $T'U'V'$ are equal. Since V corresponds to V', the measure of angle V' is equal to the measure of angle V. The figure shows that the measure of angle V is $25°$, so it follows that the measure of angle V' is also $25°$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length of $\overline{VU}$, not the measure, in degrees, of angle V'.

▼ 2.11 — B

Choice B is correct. In the figure shown, the angle measuring $y°$ is congruent to its vertical angle formed by lines s and m, so the measure of the vertical angle is also $y°$. The vertical angle forms a same-side interior angle pair with the angle measuring $x°$. It's given that lines r and s are parallel. Therefore, same-side interior angles in the figure are supplementary, which means the sum of the measure of the vertical angle and the measure of the angle measuring $x°$ is $180°$, or $x + y = 180$. Subtracting x from both sides of this equation yields $y = 180 - x$. Substituting $180 - x$ for y in the inequality $y < 65$ yields $180 - x < 65$. Adding x to both sides of this inequality yields $180 < 65 + x$. Subtracting 65 from both sides of this inequality yields $115 < x$, or $x > 115$. Thus, if $y < 65$, it must be true that $x > 115$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $x + y$ must be equal to, not less than, 180.

Choice D is incorrect. $x + y$ must be equal to, not greater than, 180.

▼ 2.12 — B

Choice B is correct. It's given that quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R, and S correspond to P', Q', R', and S', respectively. Since corresponding angles of similar quadrilaterals are congruent, it follows that the measure of angle P is equal to the measure of angle P'. It's given that the measure of angle P is $30°$. Therefore, the measure of angle P' is $30°$.

Choice A is incorrect. This is $\frac{1}{3}$ the measure of angle P'.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is 3 times the measure of angle P'.

▼ 2.13 — D

Choice D is correct. It's given that triangle *ABC* is similar to triangle *XYZ*, such that *A*, *B*, and *C* correspond to *X*, *Y*, and *Z*, respectively. Therefore, side *AB* corresponds to side *XY*. Since the length of each side of triangle *XYZ* is 2 times the length of its corresponding side in triangle *ABC*, it follows that the measure of side *XY* is 2 times the measure of side *AB*. Thus, since the measure of side *AB* is 16, then the measure of side *XY* is 216, or 32.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure of side *AB*, not side *XY*.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.14 — A

Choice A is correct. It's given that each side of equilateral triangle S is multiplied by a scale factor of *k* to create equilateral triangle T. Since the length of each side of triangle T is greater than the length of each side of triangle S, the scale factor of *k* must be greater than 1. Of the given choices, only $\frac{29}{28}$ is greater than 1.

Choice B is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of 1, the length of each side of triangle T would be equal to the length of each side of triangle S.

Choice C is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of $\frac{28}{29}$, the length of each side of triangle T would be less than the length of each side of triangle S.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.15 — C

Choice C is correct. It's given that the sum of the measures of two angles in a triangle is $64°$. The sum of the measures of all three angles in a triangle is $180°$. It follows that the measure of the third angle is the difference between $180°$ and $64°$, or $180 - 64$, which is $116°$. Therefore, the measure of the third angle is $116°$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.16 — B

Choice B is correct. The figure shows transversal k intersecting lines r and s, where the angles labeled $w°$ and $y°$ are two corresponding angles and the angles labeled $x°$ and $z°$ are two corresponding angles. When a transversal intersects two lines, the lines are parallel if and only if two corresponding angles are congruent. It's given that $w = 147$. If $y = 147$, then the two corresponding angles labeled $w°$ and $y°$ are congruent, which is sufficient to prove that lines r and s are parallel.

Choice A is incorrect. The angle labeled $x°$ forms a linear pair with the angle labeled $w°$, so $x = 180 - 147$, or $x = 33$, regardless of whether lines r and s are parallel.

Choice C is incorrect. Substituting 147 for w in this equation yields $147 + y = 180$, or $y = 33$. Since the angle labeled $w°$ and the angle labeled $y°$ are corresponding angles, if w and y aren't equal, then lines r and s can't be parallel.

Choice D is incorrect. The relationship $y + z = 180$ describes a linear pair at line s, which is true regardless of whether lines r and s are parallel.

▼ 2.17 — 84

The correct answer is 84. The sum of the measures of the interior angles of a triangle is $180°$. It's given that in triangle JKL, the measures of $\angle K$ and $\angle L$ are each $48°$. Adding the measures, in degrees, of $\angle K$ and $\angle L$ gives $48 + 48$, or 96. Therefore, the measure of $\angle J$, in degrees, is $180 - 96$, or 84.

▼ 2.18 — C

Choice C is correct. It's given that $CD = EF$ and $FC = DE$, and the figure shows that $\overline{DF}$ is a side of both $\triangle CDF$ and $\triangle EFD$. Since $\triangle CDF$ and $\triangle EFD$ have three pairs of congruent sides, the triangles are congruent, where $\angle CFD$ corresponds to $\angle EDF$ and $\angle CDF$ corresponds to $\angle EFD$. It's given that the measure of $\angle EDF$ is $65°$ and the measure of $\angle EFD$ is $30°$. Since corresponding angles of congruent triangles are congruent, $\angle CFD$ is congruent to $\angle EDF$ and $\angle CDF$ is congruent to $\angle EFD$. Therefore, the measure of $\angle CFD$ is $65°$ and the measure of $\angle CDF$ is $30°$. Since $\angle CDE$ is made up of $\angle CDF$ and $\angle EDF$, it follows that the measure of $\angle CDE$ is equal to the sum of the measures of $\angle CDF$ and $\angle EDF$. Therefore, the measure of $\angle CDE$ is equal to $(30 + 65)°$, or $95°$.

Choice A is incorrect. This is the measure of $\angle CDF$, not the measure of $\angle CDE$.

Choice B is incorrect. This is the measure of $\angle CFD$, not the measure of $\angle CDE$.

Choice D is incorrect. This is the sum of the measures of $\angle CFD$ and $\angle EDF$, not $\angle CDF$ and $\angle EDF$.

▼ 2.19 — 54

The correct answer is 54. It's given that $AB = BC$, so triangle ABC is isosceles. It's also given that line segments DE and AC are parallel. Since line segments DE and AC are parallel, triangle BDE is similar to triangle BAC, which means triangle BDE is also isosceles. It's given that point O is the midpoint of line segment DE. Since triangle BDE is isosceles, line segment BO splits triangle BDE into two congruent triangles, where $\angle BOD$ is a right angle. Therefore, $\angle BFA$ is also a right angle. The figure shows that the measure of $\angle OAF$ is $30°$. It follows that triangle AOF is a 30-60-90 triangle, where the measure of $\angle AOF$ is $60°$. In a 30-60-90 triangle, the side opposite the $30°$ angle is half the length of the side opposite the $90°$ angle. Let x represent the length of line segment OF and let $2x$ represent the length of line segment AO. It's given that $AO = BO$. Since $AO = BO$, the length of line segment BO is also $2x$. It follows that the length of line segment BF is $BO + OF$, which is $2x + x$, or $3x$. Since triangle BDE is similar to triangle BAC, the ratio of corresponding side lengths is equal to the ratio of the heights of the triangles. This means that $\frac{BE}{BC} = \frac{BO}{BF}$. Substituting $2x$ for BO and $3x$ for BF in this equation yields $\frac{BE}{BC} = \frac{2x}{3x}$, or $\frac{BE}{BC} = \frac{2}{3}$. It's given that $AB = 81$ and $AB = BC$, so $BC = 81$. Substituting 81 for BC in the equation $\frac{BE}{BC} = \frac{2}{3}$ yields $\frac{BE}{81} = \frac{2}{3}$. Multiplying both sides of this equation by 81 yields $BE = \frac{2}{3}(81)$, or $BE = 54$.

▼ 2.20 — C

Choice C is correct. It's given that the measure of angle A is $52°$ and the measure of angle P is $52°$. Therefore, angle A is congruent to angle P. Choice C states that the measures of angle B and angle R are $32°$ and $96°$, respectively. The triangle sum theorem states that the measures of the angles in a triangle sum to $180°$. In triangle ABC, if the measure of angle B is $32°$, it follows that the measure of angle C is $(180 - (52 + 32))°$, or $96°$. In triangle PQR, if the measure of angle R is $96°$, it follows that the measure of angle Q is $(180 - (52 + 96))°$, or $32°$. Therefore, in triangles ABC and PQR, angle B is congruent to angle Q, and angle C is congruent to angle R. If two triangles have three pairs of congruent angles, then the triangles are similar. Therefore, the additional piece of information that is sufficient to prove

that triangle *ABC* is similar to triangle *PQR* is that the measures of angle *B* and angle *R* are $32°$ and $96°$, respectively.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. If the measure of angle *B* is $52°$, it follows that the measure of angle *C* is $(180 - (52 + 52))°$, or $76°$. If the measure of angle *Q* is $32°$, which isn't equal to any angle measure in triangle *ABC*, the triangles can't be similar.

▼ 2.21 — 156

The correct answer is 156. In the figure shown, the sum of the measures of angle *UVS* and angle *RVS* is $180°$. It's given that the measure of angle *RVS* is $41°$. Therefore, the measure of angle *UVS* is $180 - 41°$, or $139°$. The sum of the measures of the interior angles of a triangle is $180°$. In triangle *UVS*, the measure of angle *UVS* is $139°$ and it's given that the measure of angle *VST* is $29°$. Thus, the measure of angle *VUS* is $180 - 139 - 29°$, or $12°$. It's given that $RT = TU$. Therefore, triangle *TUR* is an isosceles triangle and the measure of *VUS* is equal to the measure of angle *TRU*. In triangle *TUR*, the measure of angle *VUS* is $12°$ and the measure of angle *TRU* is $12°$. Thus, the measure of angle *UTR* is $180 - 12 - 12°$, or $156°$. The figure shows that the measure of angle *UTR* is $x°$, so the value of x is 156.

▼ 2.22 — C

Choice C is correct. It's given that triangle *XYZ* is similar to triangle *RST*, such that *X*, *Y*, and *Z* correspond to *R*, *S*, and *T*, respectively. Since corresponding angles of similar triangles are congruent, it follows that the measure of $\angle Z$ is congruent to the measure of $\angle T$. It's given that the measure of $\angle Z$ is $20°$. Therefore, the measure of $\angle T$ is $20°$.

Choice A is incorrect and may result from a conceptual error.

Choice B is incorrect. This is half the measure of $\angle Z$.

Choice D is incorrect. This is twice the measure of $\angle Z$.

▼ 2.23 — 14.66, 14.67, 44/3

The correct answer is $\frac{44}{3}$. It's given that in triangle *RST*, angle *T* is a right angle. The area of a right triangle can be found using the formula $A = \frac{1}{2} l_1 l_2$, where *A* represents the area of the right triangle, l_1 represents the length of one leg of the triangle, and l_2 represents the length

of the other leg of the triangle. In triangle RST, the two legs are $\overline{RT}$ and $\overline{ST}$. Therefore, if the length of $\overline{RT}$ is 72 and the area of triangle RST is 792, then $792 = \frac{1}{2}72ST$, or $792 = 36ST$. Dividing both sides of this equation by 36 yields $22 = ST$. Therefore, the length of $\overline{ST}$ is 22. It's also given that point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. It follows that angle LKS is a right angle. Since triangles RST and LSK share angle S and have right angles T and K, respectively, triangles RST and LSK are similar triangles. Therefore, the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is equal to the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$. If the length of $\overline{RT}$ is 72 and the length of $\overline{LK}$ is 24, it follows that the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is $\frac{72}{24}$, or 3, so the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$ is 3. Therefore, $\frac{22}{SK} = 3$. Multiplying both sides of this equation by SK yields $22 = 3SK$. Dividing both sides of this equation by 3 yields $\frac{22}{3} = SK$. Since the length of $\overline{ST}$, 22, is the sum of the length of $\overline{SK}$, $\frac{22}{3}$, and the length of $\overline{KT}$, it follows that the length of $\overline{KT}$ is $22 - \frac{22}{3}$, or $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

▼ 2.24 — ?

The correct answer is 4.5. According to the properties of right triangles, BD divides triangle ABC into two similar triangles, ABD and BCD. The corresponding sides of ABD and BCD are proportional, so the ratio of BD to AD is the same as the ratio of DC to BD. Expressing this information as a proportion gives $\frac{6}{8} = \frac{DC}{6}$. Solving the proportion for DC results in $DC = 4.5$. Note that 4.5 and 9/2 are examples of ways to enter a correct answer.

▼ 2.25 — 83

The correct answer is 83. It's given that in the figure, $AC = CD$. Thus, triangle ACD is an isosceles triangle and the measure of angle CDA is equal to the measure of angle CAD. The sum of the measures of the interior angles of a triangle is $180°$. Thus, the sum of the measures of the interior angles of triangle ACD is $180°$. It's given that the measure of angle ACD is $104°$. It follows that the sum of the measures of angles CDA and CAD is $180 - 104°$, or $76°$. Since the measure of angle CDA is equal to the measure of angle CAD, the measure of angle CDA is half of $76°$, or $38°$. The sum of the measures of the interior angles of triangle BDE is $180°$. It's given that the measure of angle EBC is $45°$. Since the measure of angle BDE, which is the same angle as angle CDA, is $38°$, it follows that the measure of angle DEB is $180 - 45 - 38°$, or $97°$. Since angle DEB and angle AEB form a straight line, the sum of the measures of these angles is $180°$. It's given in the figure that the measure of angle AEB is $x°$. It follows that $97 + x = 180$. Subtracting 97 from both sides of this equation yields $x = 83$.

▼ 2.26 — D

Choice D is correct. In the figure, it's given that segment *AD* is perpendicular to line *l* at point *A*. Statement I is that segment *AD* is perpendicular to line *m* at point *D*. It follows that angles *BAC* and *EDC* are right angles. Therefore, triangles *ABC* and *DEC* are right triangles. At least one pair of congruent corresponding sides is needed to determine triangle congruence by the side-angle-side (SAS) condition or the angle-side-angle (ASA) condition, so statement I is not sufficient to determine that triangle *ABC* is congruent to triangle *DEC*. Statement II is that $AC = 13$ and $DC = 13$. It follows that triangles *ABC* and *DEC* share one pair of congruent corresponding sides. Two additional pairs of congruent corresponding parts are needed to determine triangle congruence by SAS condition, ASA condition, or side-side-side (SSS) condition, so statement II is not sufficient to determine that triangle *ABC* is congruent to triangle *DEC*. Therefore, neither I nor II is sufficient by itself to determine that triangle *ABC* is congruent to triangle *DEC*.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 2.27 — B

Choice B is correct. It's given that in triangles *ABC* and *FGH*, angles *B* and *G* each measure $51°$, the length of $\overline{AB}$ is 6, and the length of $\overline{FG}$ is 18. For statement I, if the measure of angle *A* is equal to the measure of angle *F*, then triangles *ABC* and *FGH* have two pairs of congruent angles, angles *A* and *F* and angles *B* and *G*. When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement I is sufficient. For statement II, if the measure of angle *C* is equal to the measure of angle *H*, then triangles *ABC* and *FGH* have 2 pairs of congruent angles, angles *C* and *H* and angles *B* and *G*. When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement II is sufficient. For statement III, if the length of $\overline{FH}$ is 3 times the length of $\overline{AC}$, then the ratio of the length of $\overline{FH}$ to the length of $\overline{AC}$ is 3. It's given that the length of $\overline{FG}$ is 18 and the length of $\overline{AB}$ is 6. Therefore, the ratio of the length of $\overline{FG}$ to the length of $\overline{AB}$ is also 3, so there are two pairs of sides in proportion. However, angle *B* isn't between sides *AB* and *AC*, and angle *G* isn't between sides *FG* and *FH*. Since the given angle measures aren't between the proportional sides, this information isn't sufficient to prove that the triangles are similar. Therefore, statements I and II are each sufficient, but statement III isn't sufficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.28 — 153

The correct answer is 153. Since it's given that $\overline{PQ}$ is parallel to $\overline{XY}$ and angle Y is a right angle, angle ZQP is also a right angle. Angle ZPQ is complementary to angle XZY, which means its measure, in degrees, is $90 - 63$, or 27. Since angle XPQ is supplementary to angle ZPQ, its measure, in degrees, is $180 - 27$, or 153.

▼ 2.29 — D

Choice D is correct. Corresponding angles in similar figures have equal measure. It's given that quadrilaterals $PQRS$ and $WXYZ$ are similar and that P, Q, and R correspond to W, X, and Y. It follows that $\angle S$ corresponds to $\angle Z$. It's also given that the measure of $\angle S$ is $135°$. Therefore, the measure of $\angle Z$ is $135°$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the supplement of the measure of $\angle Z$, not the measure of $\angle Z$.

▼ 2.30 — 54

The correct answer is 54. The figure shown includes two triangles, triangle WQX and triangle YQZ, such that angle WQX and angle YQZ are vertical angles. It follows that angle WQX is congruent to angle YQZ. It's also given in the figure that the measures of angle W and angle Y are $a°$. Therefore angle W is congruent to angle Y. Since triangle WQX and triangle YQZ have two pairs of congruent angles, triangle WQX is similar to triangle YQZ by the angle-angle similarity postulate, where $\overline{YZ}$ corresponds to $\overline{WX}$, and $\overline{YQ}$ corresponds to $\overline{WQ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{WX} = \frac{YQ}{WQ}$. It's given that $YQ = 63$, $WQ = 70$, and $WX = 60$. Substituting 63 for YQ, 70 for WQ, and 60 for WX in the equation $\frac{YZ}{WX} = \frac{YQ}{WQ}$ yields $\frac{YZ}{60} = \frac{63}{70}$. Multiplying each side of this equation by 60 yields $YZ = \frac{63}{70}60$, or $YZ = 54$. Therefore, the length of $\overline{YZ}$ is 54.

▼ 2.31 — 480

The correct answer is 480. It's given in the figure that angle *ACB* and angle *AED* are right angles. It follows that angle *ACB* is congruent to angle *AED*. It's also given that angle *BAC* and angle *DAE* are the same angle. It follows that angle *BAC* is congruent to angle *DAE*. Since triangles *ABC* and *ADE* have two pairs of congruent angles, the triangles are similar. Sides *AB* and *AC* in triangle *ABC* correspond to sides *AD* and *AE*, respectively, in triangle *ADE*. Corresponding sides in similar triangles are proportional. Therefore, $\frac{AD}{AB} = \frac{AE}{AC}$. It's given that $AC = 3$ units and $CE = 21$ units. Therefore, $AE = 24$ units. It's also given that $AB = \sqrt{34}$ units. Substituting 3 for *AC*, 24 for *AE*, and $\sqrt{34}$ for *AB* in the equation $\frac{AD}{AB} = \frac{AE}{AC}$ yields $\frac{AD}{\sqrt{34}} = \frac{24}{3}$, or $\frac{AD}{\sqrt{34}} = 8$. Multiplying each side of this equation by $\sqrt{34}$ yields $AD = 8\sqrt{34}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length *c* and legs with lengths *a* and *b*, then $a^2 + b^2 = c^2$. Since triangle *ADE* is a right triangle, it follows that *AD* represents the length of the hypotenuse, *c*, and *DE* and *AE* represent the lengths of the legs, *a* and *b*. Substituting 24 for *b* and $8\sqrt{34}$ for *c* in the equation $a^2 + b^2 = c^2$ yields $a^2 + 24^2 = 8\sqrt{34}^2$, which is equivalent to $a^2 + 576 = 6434$, or $a^2 + 576 = 2{,}176$. Subtracting 576 from both sides of this equation yields $a^2 = 1{,}600$. Taking the square root of both sides of this equation yields $a = \pm 40$. Since *a* represents a length, which must be positive, the value of *a* is 40. Therefore, $DE = 40$. Since *DE* and *AE* represent the lengths of the legs of triangle *ADE*, it follows that *DE* and *AE* can be used to calculate the area, in square units, of the triangle as $\frac{1}{2}4024$, or 480. Therefore, the area, in square units, of triangle *ADE* is 480.

▼ 2.32 — C

Choice C is correct. First, $\overline{MP}$ is the hypotenuse of right $\triangle MNP$, whose legs have lengths 3 and 4. Therefore, $(MP)^2 = 3^2 + 4^2$, so $(MP)^2 = 25$ and $MP = 5$. Second, because $\angle MNP$ corresponds to $\angle NQP$ and because $\angle MPN$ corresponds to $\angle NPQ$, $\triangle MNP$ is similar to $\triangle NQP$. The ratio of corresponding sides of similar triangles is constant, so $\frac{NQ}{MN} = \frac{NP}{MP}$. Since $MP = 5$ and it's given that $MN = 3$ and $NP = 4$, $\frac{NQ}{3} = \frac{4}{5}$. Solving for NQ results in $NQ = \frac{12}{5}$, or 2.4.

Choices A, B, and D are incorrect and may result from setting up incorrect ratios.

▼ 2.33 — C

Choice C is correct. Since angles *B* and *E* each have the same measure and angles *C* and *F* each have the same measure, triangles *ABC* and *DEF* are similar, where side *BC* corresponds to side *EF*. To determine whether two similar triangles are congruent, it is sufficient to determine whether one pair of corresponding sides are congruent. Therefore, to determine whether triangles *ABC* and *DEF* are congruent, it is sufficient to determine whether sides *BC* and *EF* have equal length. Thus, the lengths of *BC* and *EF* are sufficient to determine whether triangle *ABC* is congruent to triangle *DEF*.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. The given information is sufficient to determine that triangles *ABC* and *DEF* are similar, but not whether they are congruent.

▼ 2.34 — ?

The correct answer is 30. It is given that the measure of $\angle QPR$ is $60°$. Angle MPR and $\angle QPR$ are collinear and therefore are supplementary angles. This means that the sum of the two angle measures is $180°$, and so the measure of $\angle MPR$ is $120°$. The sum of the angles in a triangle is $180°$. Subtracting the measure of $\angle MPR$ from $180°$ yields the sum of the other angles in the triangle MPR. Since $180-120=60$, the sum of the measures of $\angle QMR$ and $\angle NRM$ is $60°$. It is given that $MP=PR$, so it follows that triangle MPR is isosceles. Therefore $\angle QMR$ and $\angle NRM$ must be congruent. Since the sum of the measure of these two angles is $60°$, it follows that the measure of each angle is $30°$.

An alternate approach would be to use the exterior angle theorem, noting that the measure of $\angle QPR$ is equal to the sum of the measures of $\angle QMR$ and $\angle NRM$. Since both angles are equal, each of them has a measure of $30°$.

▼ 2.35 — A

Choice A is correct. It's given that *WZ* and *XY* intersect at point *Q*. It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of $a°$, so they are congruent to each other. Therefore, $\triangle QWX$ and $\triangle QYZ$ are similar triangles, where *WQ* corresponds to *YQ* and *WX* corresponds to *YZ*. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ}=\frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$.

Substituting 21 for YQ, 70 for WQ, and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of YZ is 18.

Choice B is incorrect. This is the length of QZ, not YZ.

Choice C is incorrect. This is the length of XQ, not YZ.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.36 — C

Choice C is correct. Since $P = 4, 5$ and $Q = 4, 7$, side PQ is parallel to the y-axis and has a length of 2. Since $P = 4, 5$ and $R = 6, 5$, side PR is parallel to the x-axis and has a length of 2. Therefore, triangle PQR is a right isosceles triangle, where $\angle P$ has measure $90°$ and $\angle Q$ and $\angle R$ each have measure $45°$. It follows that if the measure of $\angle Q$ is $t°$, then $t = 45$. Since $L = 4, 5$ and $M = 4, 7 + k$, side LM is parallel to the y-axis and has a length of $k + 2$. Since $L = 4, 5$ and $N = 6 + k, 5$, side LN is parallel to the x-axis and has a length of $k + 2$. Therefore, triangle LMN is a right isosceles triangle, where $\angle L$ has measure $90°$ and $\angle M$ and $\angle N$ each have measure $45°$. Of the given choices, only $90 - t°$ is equal to $45°$, so the measure of $\angle N$ is $(90 - t)°$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.37 — B

Choice B is correct. In the figure, $\bar{AE}$ intersects line l and $\bar{EG}$, and alternate interior angles are both marked $x°$. It follows that $\bar{EG}$ is parallel to line l. It's given that line l is parallel to line j, so $\bar{EG}$ is also parallel to line j. Thus, the following angles also have measure $x°$: $\angle EFA$, $\angle ACD$, $\angle BAD$, and $\angle GFD$. Triangles AEF and ACD each have two angles with measure $x°$, so triangles AEF and ACD are similar. It's also given that $EF = 0.5CD$, so $AF = 0.5AD$. It follows that $AF = FD$. Triangles DFG and DAB each have one angle with measure $x°$ and share $\angle FDG$, so triangles DFG and DAB are similar. Since $AF = FD$, it follows that $AD = 2FD$, so $AB = 2FG$. Substituting $2FG$ for AB in statement I yields $2FG = 2.25EF$. It's given that $FG = 2.25CD$. Substituting $2.25CD$ for FG and $0.5CD$ for EF in the equation $2FG = 2.25EF$ yields $2(2.25CD) = 2.25(0.5CD)$, or $4.5CD = 1.125CD$. Thus,

statement I is not true. Since $AB = 2FG$, it follows that $0.5AB = FG$, or $FG = 0.5AB$. Therefore, statement II must be true. Substituting $0.5CD$ for EF, $2.25CD$ for FG, and FD for AF in statement III yields $\frac{0.5CD}{2.25CD} = \frac{FD}{FD}$, or $\frac{2}{9} = 1$. Thus, statement III is not true. Therefore, statement II only must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.38 — 101/2, 50.5

The correct answer is $\frac{101}{2}$. In the figure, lines q, r, and s form a triangle. One interior angle of this triangle is vertical to the angle marked $a°$; therefore, the interior angle also has measure $a°$. It's given that $a = 43$. Therefore, the interior angle of the triangle has measure $43°$. A second interior angle of the triangle forms a straight line, q, with the angle marked $b°$. Therefore, the sum of the measures of these two angles is $180°$. It's given that $b = 122$. Therefore, the angle marked $b°$ has measure $122°$ and the second interior angle of the triangle has measure $180 - 122°$, or $58°$. The sum of the interior angles of a triangle is $180°$. Therefore, the measure of the third interior angle of the triangle is $180 - 43 - 58°$, or $79°$. It's given that parallel lines q and t are intersected by line r. It follows that the triangle's interior angle with measure $79°$ is congruent to the same side interior angle between lines q and t formed by lines t and r. Since this angle is supplementary to the two angles marked $w°$, the sum of $79°$, $w°$, and $w°$ is $180°$. It follows that $79 + w + w = 180$, or $79 + 2w = 180$. Subtracting 79 from both sides of this equation yields $2w = 101$. Dividing both sides of this equation by 2 yields $w = \frac{101}{2}$. Note that 101/2 and 50.5 are examples of ways to enter a correct answer.

▼ 2.39 — D

Choice D is correct. It's given that in triangle ABC, the measure of angle B is $90°$ and BD is an altitude of the triangle. Therefore, the measure of angle BDC is $90°$. It follows that angle B is congruent to angle D and angle C is congruent to angle C. By the angle-angle similarity postulate, triangle ABC is similar to triangle BDC. Since triangles ABC and BDC are similar, it follows that $\frac{AC}{AB} = \frac{BC}{BD}$. It's also given that the length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. Therefore, the length of $\overline{AC}$ is $15 + 23$, or 38. Substituting 15 for AB and 38 for AC in the equation $\frac{AC}{AB} = \frac{BC}{BD}$ yields $\frac{38}{15} = \frac{BC}{BD}$. Therefore, the value of $\frac{BC}{BD}$ is $\frac{38}{15}$.

Choice A is incorrect. This is the value of $\frac{BD}{BC}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.40 — D

Choice D is correct. The figure shows that angle MRL and angle PRQ are vertical angles. Since vertical angles are congruent, angle MRL and angle PRQ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, alternate interior angles MLR and PQR are congruent. Since triangles LMR and PQR have two pairs of congruent angles, the triangles are similar. Sides LR and MR in triangle LMR correspond to sides RQ and RP, respectively, in triangle PQR. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{RQ}{LR} = \frac{RP}{MR}$. It's given that the lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. Substituting 6 for MR, 7 for LR, and 11 for RP in the equation $\frac{RQ}{LR} = \frac{RP}{MR}$ yields $\frac{RQ}{7} = \frac{11}{6}$. Multiplying each side of this equation by 7 yields $RQ = \frac{11}{6}7$, or $RQ = \frac{77}{6}$. It's given that $\overline{LQ}$ intersects $\overline{MP}$ at point R, so $LQ = LR + RQ$. Substituting 7 for LR and $\frac{77}{6}$ for RQ in this equation yields $LQ = 7 + \frac{77}{6}$, or $LQ = \frac{119}{6}$. Therefore, the length of $\overline{LQ}$ is $\frac{119}{6}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{RQ}$, not $\overline{LQ}$.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.41 — 54

The correct answer is 54. It's given that in triangle ABC, point D on side AB is connected by a line segment with point E on side AC such that line segment DE is parallel to side BC. It follows that parallel segments DE and BC are intersected by sides AB and AC. If two parallel segments are intersected by a third segment, corresponding angles are congruent. Thus, corresponding angles C and AED are congruent and corresponding angles B and ADE are congruent. Since triangle ADE has two angles that are each congruent to an angle in triangle ABC, triangle ADE is similar to triangle ABC by the angle-angle similarity postulate, where side DE corresponds to side BC, and side AE corresponds to side AC. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{DE}{BC} = \frac{AE}{AC}$. Since point E lies on side AC, $AE + CE = AC$. It's given that $CE = 2AE$. Substituting $2AE$ for CE in the equation $AE + CE = AC$ yields $AE + 2AE = AC$, or $3AE = AC$. It's given that

$BC = 162$. Substituting 162 for BC and $3AE$ for AC in the equation $\frac{DE}{BC} = \frac{AE}{AC}$ yields $\frac{DE}{162} = \frac{AE}{3AE}$, or $\frac{DE}{162} = \frac{1}{3}$. Multiplying both sides of this equation by 162 yields $DE = 54$. Thus, the length of line segment DE is 54.

▼ 2.42 — D

Choice D is correct. The figure shows that $\angle MRL$ and $\angle PRQ$ are vertical angles. Since vertical angles are congruent, $\angle MRL$ and $\angle PRQ$ are congruent. It's given that $\bar{LM}$ is parallel to $\bar{PQ}$. The figure also shows that $\bar{LQ}$ intersects $\bar{LM}$ and $\bar{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, $\angle MLR$ and $\angle PQR$ are congruent. Since $\triangle LMR$ and $\triangle PQR$ have two pairs of congruent angles, the triangles are similar, where $\bar{MR}$ in $\triangle LMR$ corresponds to $\bar{PR}$ in $\triangle PQR$. Since the areas of similar triangles are proportional to the squares of the lengths of corresponding sides, it follows that $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = (\frac{RP}{MR})^2$. It's given that the lengths of $\bar{MR}$ and $\bar{RP}$ are 8 and 14 units, respectively, and the area of $\triangle LMR$ is 36 square units. Substituting 14 for RP, 8 for MR, and 36 for $\text{area of } \triangle LMR$ in the equation $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = (\frac{RP}{MR})^2$ yields $\frac{\text{area of } \triangle PQR}{36} = (\frac{14}{8})^2$, which is equivalent to $\frac{\text{area of } \triangle PQR}{36} = (\frac{7}{4})^2$, or $\frac{\text{area of } \triangle PQR}{36} = \frac{49}{16}$. Multiplying each side of this equation by 36 yields $\text{area of } \triangle PQR = 36(\frac{49}{16})$, or $\text{area of } \triangle PQR = \frac{441}{4}$. Therefore, the area of $\triangle PQR$ is $\frac{441}{4}$ square units.

Choice A is incorrect. This is the result of multiplying 36 by $(\frac{8}{14})^2$ instead of by $(\frac{14}{8})^2$.

Choice B is incorrect. This is the result of multiplying 36 by $\frac{8}{14}$ instead of by $(\frac{14}{8})^2$.

Choice C is incorrect. This is the result of multiplying 36 by $\frac{14}{8}$ instead of by $(\frac{14}{8})^2$.

▼ 2.43 — C

Choice C is correct. Since $RT = TU$, it follows that $\triangle RTU$ is an isosceles triangle with base RU. Therefore, $\angle TRU$ and $\angle TUR$ are the base angles of an isosceles triangle and are congruent. Let the measures of both $\angle TRU$ and $\angle TUR$ be $t°$. According to the triangle sum theorem, the sum of the measures of the three angles of a triangle is $180°$. Therefore, $114° + 2t° = 180°$, so $t = 33$.

Note that $\angle TUR$ is the same angle as $\angle SUV$. Thus, the measure of $\angle SUV$ is $33°$. According to the triangle exterior angle theorem, an external angle of a triangle is equal to the sum of the opposite interior angles. Therefore, $x°$ is equal to the sum of the measures of $\angle VSU$ and $\angle SUV$; that is, $31° + 33° = 64°$. Thus, the value of x is 64.

Choice B is incorrect. This is the measure of $\angle STR$, but $\angle STR$ is not congruent to $\angle SVR$. Choices A and D are incorrect and may result from a calculation error.

▼ 2.44 — A

Choice A is correct. It's given that two lines intersect at exactly one point, forming two acute angles and two obtuse angles. When two lines intersect, opposite angles are congruent and adjacent angles are supplementary. It's given that the measure of one of the angles is $(9x - 140)°$, so the measure of the opposite angle is $(9x - 140)°$, and the measure of each adjacent angle is $(180 - (9x - 140))°$, or $(-9x + 320)°$. The possible sums of the measures of any two of the angles are $(9x - 140)° + (9x - 140)°$, or $(18x - 280)°$, $(-9x + 320)° + (-9x + 320)°$, or $(-18x + 640)°$, and $(9x - 140)° + (-9x + 320)°$, or $180.°$ Of the given choices, only $(-18x + 280)°$ is not equivalent to any of these expressions and could not be the sum of the measures of any two of the angles.

Choice B is incorrect. This is the sum of the measures of two angles that are adjacent to the given angle.

Choice C is incorrect. This is the sum of the measure of the given angle and the measure of its opposite angle.

Choice D is incorrect. This is the sum of the measure of the given angle and the measure of an angle adjacent to the given angle.

▼ 2.45 — 123

The correct answer is 123. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180 degrees. It's given that the measure of $\angle SQX$ is $48°$ and the measure of $\angle SXQ$ is $86°$. Since points S, Q, and X form a triangle, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle QSX$ is $180 - 48 - 86$, or 46. It's also given that the measure of $\angle SWU$ is $85°$. Since $\angle SWU$ and $\angle SWR$ are supplementary angles, the sum of their measures is 180 degrees. It follows that the measure, in degrees, of $\angle SWR$ is $180 - 85$, or 95. Since points R, S, and W form a triangle, and $\angle RSW$ is the same angle as $\angle QSX$, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle WRS$ is $180 - 46 - 95$, or 39. It's given that the measure of $\angle VTU$ is $162°$. Since $\angle VTU$ and $\angle STU$ are supplementary angles, the sum of their measures is 180 degrees. It follows that the measure, in degrees, of $\angle STU$ is $180 - 162$

, or 18. Since points *R*, *T*, and *U* form a triangle, and ∠*URT* is the same angle as ∠*WRS*, it follows from the triangle angle sum theorem that the measure, in degrees, of ∠*TUR* is 180 - 39 - 18, or 123.

▼ 2.46 — ?

The correct answer is either $\frac{10}{3}$, $\frac{15}{4}$, or $\frac{25}{6}$. The area of triangle ABC can be expressed as $\frac{1}{2}(5+7)y$ or $6y$. It's given that the area of triangle ABC is at least 48 but no more than 60. It follows that $48 \le 6y \le 60$. Dividing by 6 to isolate y in this compound inequality yields $8 \le y \le 10$. Since y is an integer, $y = 8, 9, \text{ or } 10$. In the given figure, the two right triangles shown are similar because they have two pairs of congruent angles: their respective right angles and angle A. Therefore, the following proportion is true: $\frac{x}{y} = \frac{5}{12}$. Substituting 8 for y in the proportion results in $\frac{x}{8} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{10}{3}$. Substituting 9 for y in the proportion results in $\frac{x}{9} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{15}{4}$. Substituting 10 for y in the proportion results in $\frac{x}{10} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{25}{6}$. Note that 10/3, 15/4, 25/6, 3.333, 3.75, 4.166, and 4.167 are examples of ways to enter a correct answer.

▼ 2.47 — D

Choice D is correct. Two triangles are similar if they have three pairs of congruent corresponding angles. It's given that angles *L* and *R* each measure 60°, and so these corresponding angles are congruent. If angle *M* is 70°, then angle *N* must be 50° so that the sum of the angles in triangle *LMN* is 180°. If angle *T* is 50°, then angle *S* must be 70° so that the sum of the angles in triangle *RST* is 180°. Therefore, if the measures of angles *M* and *T* are 70° and 50°, respectively, then corresponding angles *M* and *S* are both 70°, and corresponding angles *N* and *T* are both 50°. It follows that triangles *LMN* and *RST* have three pairs of congruent corresponding angles, and so the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle *LMN* is similar to triangle *RST* is that the measures of angles *M* and *T* are 70° and 50°, respectively.

Choice A is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the

triangles are similar. However, the two sides given are not proportional and the angle given is not included by the given sides.

Choice B is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the angle given is not included between the proportional sides.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.48 — ?

The correct answer is 97. The intersecting lines form a triangle, and the angle with measure of $x°$ is an exterior angle of this triangle. The measure of an exterior angle of a triangle is equal to the sum of the measures of the two nonadjacent interior angles of the triangle. One of these angles has measure of $23°$ and the other, which is supplementary to the angle with measure $106°$, has measure of $180° - 106° = 74°$. Therefore, the value of x is $23 + 74 = 97$.

▼ 2.49 — 47

The correct answer is 47. It's given that the measure of angle *B* is 139 degrees. Therefore, the exterior angle formed by extending segment *AB* at point *B* has measure 180 - 139, or 41, degrees. It's given that segment *AB* is parallel to segment *DE*. Extending segment *BC* at point *C* and extending segment *DE* at point *D* until the two segments intersect results in a transversal that intersects two parallel line segments. One of these intersection points is point *B*, and let the other intersection point be point *X*. Since segment *AB* is parallel to segment *DE*, alternate interior angles are congruent. Angle *CXD* and the exterior angle formed by extending segment *AB* at point *B* are alternate interior angles. Therefore, the measure of angle *CXD* is 41 degrees. It's given that the measure of angle *D* in pentagon *ABCDE* is 174 degrees. Therefore, angle *CDX* has measure 180 - 174, or 6, degrees. Since angle *C* in pentagon *ABCDE* is an exterior angle of triangle *CDX*, it follows that the measure of angle *C* is the sum of the measures of angles *CDX* and *CXD*. Therefore, the measure, in degrees, of angle *C* is 6 + 41, or 47.

Alternate approach: A line can be created that's perpendicular to segments *AB* and *DE* and passes through point *C*. Extending segments *AB* and *DE* at points *B* and *D*, respectively, until they intersect this line yields two right triangles. Let these intersection points be point *X* and point *Y*, and the two right triangles be triangle *BXC* and triangle *DYC*. It's given that the measure of angle *B* is 139 degrees. Therefore, angle *CBX* has measure 180 - 139, or 41,

degrees. Since the measure of angle *CBX* is 41 degrees and the measure of angle *BXC* is 90 degrees, it follows that the measure of angle *XCB* is $180 - 90 - 41$, or 49, degrees. It's given that the measure of angle *D* is 174 degrees. Therefore, angle *YDC* has measure $180 - 174$, or 6, degrees. Since the measure of angle *YDC* is 6 degrees and the measure of angle *CYD* is 90 degrees, it follows that the measure of angle *DCY* is $180 - 90 - 6$, or 84, degrees. Since angles *XCB*, *DCY*, and angle *C* in pentagon *ABCDE* form segment *XY*, it follows that the sum of the measures of those angles is 180 degrees. Therefore, the measure, in degrees, of angle *C* is $180 - 49 - 84$, or 47.

▼ 2.50 — 1660

The correct answer is $1{,}660$. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. When two parallel lines are intersected by a transversal line, the angles formed have the following properties: two adjacent angles are supplementary, and alternate interior angles are congruent. Therefore, each of the four acute angles have the same measure, and each of the four obtuse angles have the same measure. It's also given that the measure of one of the acute angles is $9x - 560°$. If two angles are supplementary, then the sum of their measures is $180°$. Therefore, the measure of the obtuse angle adjacent to any of the acute angles is $180 - (9x - 560)°$, or $180 - 9x + 560°$, which is equivalent to $-9x + 740°$. It's given that the sum of the measures of one of the acute angles and three of the obtuse angles is $-18x + w°$. It follows that $(9x - 560) + 3(-9x + 740) = -18x + w$, which is equivalent to $9x - 560 - 27x + 2{,}220 = -18x + w$, or $-18x + 1{,}660 = -18x + w$. Adding $18x$ to both sides of this equation yields $1{,}660 = w$.

▼ 2.51 — A

Choice A is correct. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. Since there are two parallel lines intersected by a transversal, all four acute angles have the same measure and all four obtuse angles have the same measure. Additionally, each acute angle is supplementary to each obtuse angle. It's given that the measure of one of these eight angles is $(7x - 250)°$. It follows that a supplementary angle has measure $(180 - (7x - 250))°$, or $(-7x + 430)°$. It's also given that the sum of the measures of four of the eight angles is $k°$. It follows that the possible values of k are $4(7x - 250)$; $(7x - 250) + 3(-7x + 430)$; $2(7x - 250) + 2(-7x + 430)$; $3(7x - 250) + (-7x + 430)$; and $4(-7x + 430)$. These values are equivalent to $28x - 1{,}000$; $-14x + 1{,}040$; 360; $14x - 320$; and $-28x + 1{,}720$, respectively. It follows that of the given choices, only $-14x + 1{,}540$ could NOT be equivalent to k, for all values of x.

Choice B is incorrect. This is the sum of three angles with measure $(7x - 250)°$ and one angle with measure $(-7x + 430)°$.

Choice C is incorrect. This is the sum of four angles with measure $(-7x + 430)°$.

Choice D is incorrect. This is the sum of two angles with measure $(7x - 250)°$ and two angles with measure $(-7x + 430)°$.